

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11: REPRODUCTION IN ANIMALS (SUB-STRAND 3.1) FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use what you observed and learned in Lessons 1 to 7 to write your Final Explanation. Answer each prompt in the space below it, using full sentences. Support every claim with evidence from our lessons: the frogs near Kisumu and Lake Victoria wetlands, the hens, the cows, the dissected small mammal, the sperm and ovum slides, the hormone graphs, the animations, the case studies and the debate. Use the correct scientific terms. You may add labelled drawings or flowcharts where they help your explanation. Write in your own words.

Part 1: Frogs, Hens and Cows - Different Strategies, One Goal

Frogs near Kisumu release thousands of jelly-covered eggs into water, hens lay hard-shelled eggs on land, and cows give birth to one live calf. (a) Name the type of fertilisation each animal uses and say where it happens. (b) Classify the hen, the cow, a fish or frog, and a snake as oviparous, viviparous or ovoviviparous, and say how the groups differ. (c) Use evidence from Lesson 1 to explain why frogs need thousands of eggs while a cow produces one calf.

(a) Frogs (and fish) use external fertilisation: the eggs and sperm meet in the water outside the body. Hens and cows use internal fertilisation: the sperm meets the egg inside the female's body. (b) The hen is oviparous: it lays hard-shelled eggs and the young develop in the egg outside the mother. Frogs and fish are also oviparous: they lay eggs (the frog's are jelly-covered) in water, and fertilisation is external. The cow is viviparous: the young develop inside the mother and she gives birth to a live calf. The snake is ovoviviparous: the eggs are kept inside the mother's body and hatch there, so the young are born alive, but they are not nourished by the mother as a calf is. (c) In external fertilisation, eggs and sperm are released into open water in the Lake Victoria wetlands. Many eggs are not fertilised, are eaten by predators or are washed away, and there is little or no parental care. So the frog needs to release thousands of eggs so that a few offspring survive. In the cow, fertilisation happens inside the body, so the gametes are protected and meet reliably. The embryo is protected and fed during months of internal development and the mother cares for the calf after birth. Few losses occur, so one calf is enough. External fertilisation needs water and gives many eggs, while internal fertilisation on land protects fewer offspring.

Part 2: Structure and Function in Mammals

In Lessons 2 and 3 you dissected a small mammal and studied sperm and ovum slides. (a) Name the parts of the female reproductive system you

(a) In the female we identified the ovaries, oviducts, uterus, cervix and vagina. The ovaries produce the gametes (ova). The oviduct carries the ovum and is where fertilisation occurs. The uterus is muscular and has a rich blood supply, so it supports the developing embryo. (b) Sperm: the tail (flagellum) lets it swim towards the ovum, and the many mitochondria in the middle section supply the energy for movement. The acrosome at the head holds enzymes that help it get through the ovum's protective layers to fuse with it. Ovum: it is large and has food reserves to feed the early embryo, and it is surrounded by a protective outer layer that

identified and state the job of the ovaries, the oviduct and the uterus. (b) Describe two structural features of a sperm cell and two of an ovum, and explain how each helps fertilisation. (c) Explain how these structures help a cow, or a human, produce one well-protected offspring rather than thousands of eggs.	protects it. (c) In mammals the testes and ovaries make gametes, the tubes carry them, and fertilisation happens safely inside the oviduct, so the gametes are not lost to the environment as in frog spawn. The embryo then develops in the uterus, which is muscular and well supplied with blood. Because the gametes and embryo are protected inside the body for months, only a few offspring are needed, unlike the frog's thousands of eggs in the open water. Humans make very few ova compared to the numbers of eggs a frog makes.
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Part 3: Hormones Control Release and Pregnancy

(a) Using the hormone graphs from Lesson 4, describe how FSH, LH, oestrogen and progesterone change during a 28-day menstrual cycle and what each does. (b) Explain why a woman usually releases one egg per month. Use the idea of feedback in your answer. (c) Describe what happens to hormones and body structures if fertilisation occurs, including the roles of hCG, the placenta, oxytocin and prolactin.	(a) FSH rises early in the cycle and stimulates a follicle in the ovary to develop. The developing follicle makes oestrogen, which rises before a sharp LH surge near Day 14. The LH surge causes ovulation shortly after the LH peak. After ovulation, progesterone rises in the second half of the cycle and keeps the uterine lining thick and ready for an embryo. If there is no pregnancy, progesterone falls before menstruation and the lining is shed. (b) The pituitary gland (FSH and LH) and the ovaries (oestrogen and progesterone) control each other by feedback. Negative feedback keeps hormone levels in balance: rising oestrogen and progesterone reduce FSH, so the stimulation that supports follicle development falls and usually only the one follicle that is furthest ahead keeps maturing. When the oestrogen level from that follicle is high for several days, it switches to positive feedback and produces the LH surge near Day 14, which triggers ovulation of that single follicle. These loops time the maturing of a single follicle and one ovulation, so usually only one egg is released at a time. Imbalances in these hormones can disrupt ovulation and the cycle. (c) If fertilisation occurs, the embryo implants in the uterine wall. The embryo makes the hormone hCG, which keeps the corpus luteum making progesterone so the lining is not lost. The placenta then forms. It exchanges oxygen, nutrients and wastes between mother and baby without the two bloods mixing; the umbilical cord only carries blood between the baby and the placenta. The placenta also makes hormones (progesterone and oestrogen) to support pregnancy. At birth, oxytocin strengthens the contractions of the uterus during labour, and prolactin brings about lactation (milk production).
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Part 4: When Reproduction Fails - IVF, Surrogacy and Sperm Banking

Some people say that IVF babies grow wholly in a test tube. (a) Using what you learned in Lesson 6, explain whether this statement is correct, and describe what happens in IVF and where the baby develops. (b) Choose TWO of the case-study couples (Nakuru, Kisumu, Eldoret, Nairobi or Mombasa) and explain which technology helps	(a) The statement is not correct. In IVF, fertilisation happens in a dish outside the body, but the embryo is then transferred to the uterus and develops there. The baby still needs the uterus, the placenta and hormones to develop. Hormone treatment is often used in IVF to help the ovaries release eggs. Humans use IVF to replace the step that fails, in a similar way to animals that have different strategies to make sure fertilisation and development succeed. (b) The couple in Nakuru have blocked fallopian tubes, so the sperm and egg cannot meet in the oviduct. IVF solves this because fertilisation takes place in a dish and the embryo is placed in the uterus, avoiding the blocked tubes. The man in Nairobi needs cancer treatment that could damage his sperm, so sperm banking lets him store sperm before treatment to use later. (Other correct pairings: absent uterus in Mombasa, surrogacy, in which another woman carries the embryo; low sperm count in Kisumu, IVF; hormonal imbalance in Eldoret, hormone treatment with IVF.) (c) Ethical issues include cost, consent, access and culture, for example whether families in rural areas can afford IVF, or the consent needed from the surrogate and the couple. Shared decisions matter because reproduction involves both partners, and informed, respectful choices based on accurate knowledge protect the health and rights of everyone involved.
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each one and why. (c) Give one ethical issue raised in our debate and explain why both parents should share informed decisions.	
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Part 5: Our Final Explanation	
Write a connected Final Explanation (one or two paragraphs) that answers the driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use the frog, hen and cow, the structure of human reproductive systems and gametes, hormones, and IVF or surrogacy or sperm banking in your answer. Then say how your thinking has changed since your first model in Lesson 1.	Animals ensure successful reproduction using different strategies that suit their environment. Frogs and fish living in water use external fertilisation and release thousands of eggs and sperm into water, because many are lost to predators and the environment and there is little parental care. Hens, cows and humans use internal fertilisation. The hen is oviparous and lays shelled eggs, while the cow is viviparous and bears a live calf after months inside the mother. Internal fertilisation protects the gametes and offspring, so fewer offspring are needed. In mammals, specialised structures make this work: the testes and ovaries make sperm and ova, tubes carry them, fertilisation takes place in the oviduct, and the muscular uterus supports the embryo. A sperm has a tail, mitochondria and an acrosome for movement and fusion, and the ovum has food reserves and protective layers. Hormones control the process: FSH and LH from the pituitary and oestrogen and progesterone from the ovaries work through feedback to time the release of usually one egg a month and to prepare the uterine lining. If fertilisation occurs, hCG, the placenta and progesterone sustain the pregnancy, and oxytocin and prolactin control birth and lactation. When natural reproduction fails, science can help: IVF fertilises eggs in a dish but the embryo is placed in the uterus, so babies do not grow wholly in a test tube; surrogacy helps when a woman has no uterus; and sperm banking stores sperm. These raise ethical questions about cost, consent and access, so couples should make shared, informed decisions. At the start I only compared eggs and live birth; now I can link environment, structure, hormones and biotechnology to explain the whole process.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Comparing fertilisation strategies	Correctly names external and internal fertilisation and oviparous, viviparous and ovoviviparous types, and explains the number of offspring using the site of fertilisation, losses and parental care, with frog, hen and cow evidence.	Names the types of fertilisation and gives a reasonable explanation of why frogs have more eggs than cows, with limited use of evidence.	Mixes up the types or gives only a statement that frogs lay more eggs, without a reason.
Structure and function	Accurately links reproductive organs and the structures of sperm and ovum (tail, mitochondria, acrosome, protective layers, food reserves) to their functions and to the protection of few offspring.	Names most structures and gives correct functions for some of them.	Lists some structures without functions or with several errors.
Hormonal control	Accurately describes the changes in	Describes the main hormone changes	Names a few hormones with unclear or

	FSH, LH, oestrogen and progesterone, explains single ovulation through feedback, and explains hCG, the placenta, oxytocin and prolactin in pregnancy and birth.	and their roles with minor gaps in feedback or pregnancy hormones.	incorrect roles.
Reproductive biotechnology and ethics	Explains that the baby does not grow wholly in a test tube, matches technologies to specific causes of infertility, and discusses ethical issues and shared decisions clearly.	Explains IVF, surrogacy or sperm banking correctly and mentions an ethical issue.	Describes the technologies vaguely or believes the baby grows wholly in a test tube.
Quality of the Final Explanation	A connected, well-organised explanation that answers the driving question, uses lesson evidence and scientific terms, and reflects on how the thinking has changed.	A mostly complete explanation using some evidence and terms, with limited reflection.	Disconnected or incomplete statements that do not answer the driving question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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